Distribution – General Data

1. General

Fuel is transferred from No. 1 cell to No. 4 cell and from No. 8 cell to No. 5 cell by corresponding booster pump operated ejectors to maintain maximum fuel supply in each collector cell (4 and 5). The fuel is transferred from cells 2 and 3 into 1, and from cells 6 and 7 into 8 by gravity. Each system consists of a motive line, a fuel ejector, a transfer line and a level control valve. The motive line supplies booster pump pressure to provide the pressure drop across the ejector and there by draw fuel from the cell. Fuel drawn from the cell passes to the collector cell of the tank, through the transfer line and level control valve, together with the motive line fuel. The level control valve prevents over-filling of the collector cell. Check valves are incorporated to prevent siphoning of the fuel into the main cell; the motive line check valves contain fuel filters. The crossfeed check valves have an internal bleed provided to prevent thermal pressure build-up in the engine fuel supply lines. Each engine fuel supply line contains a check valve, a filter and an emergency shut-off valve. Arrows stencilled adjacent to each check valve in the fuel manifold, indicate direction of fuel flow. This ensures correct installation of the check valves when the arrow marked on each check valve coincides with its related stencilled arrow. The right engine obtains its fuel supply from the forward tank booster pumps and the left engine from the aft tank booster pumps, but when crossfeeding is selected. both engines are fed from one tank. The fuel lines are routed from the booster pumps to No. 6 cell, through the cell, up the rear side of the wing struts, along the wings, into the engine nacelles and engines.

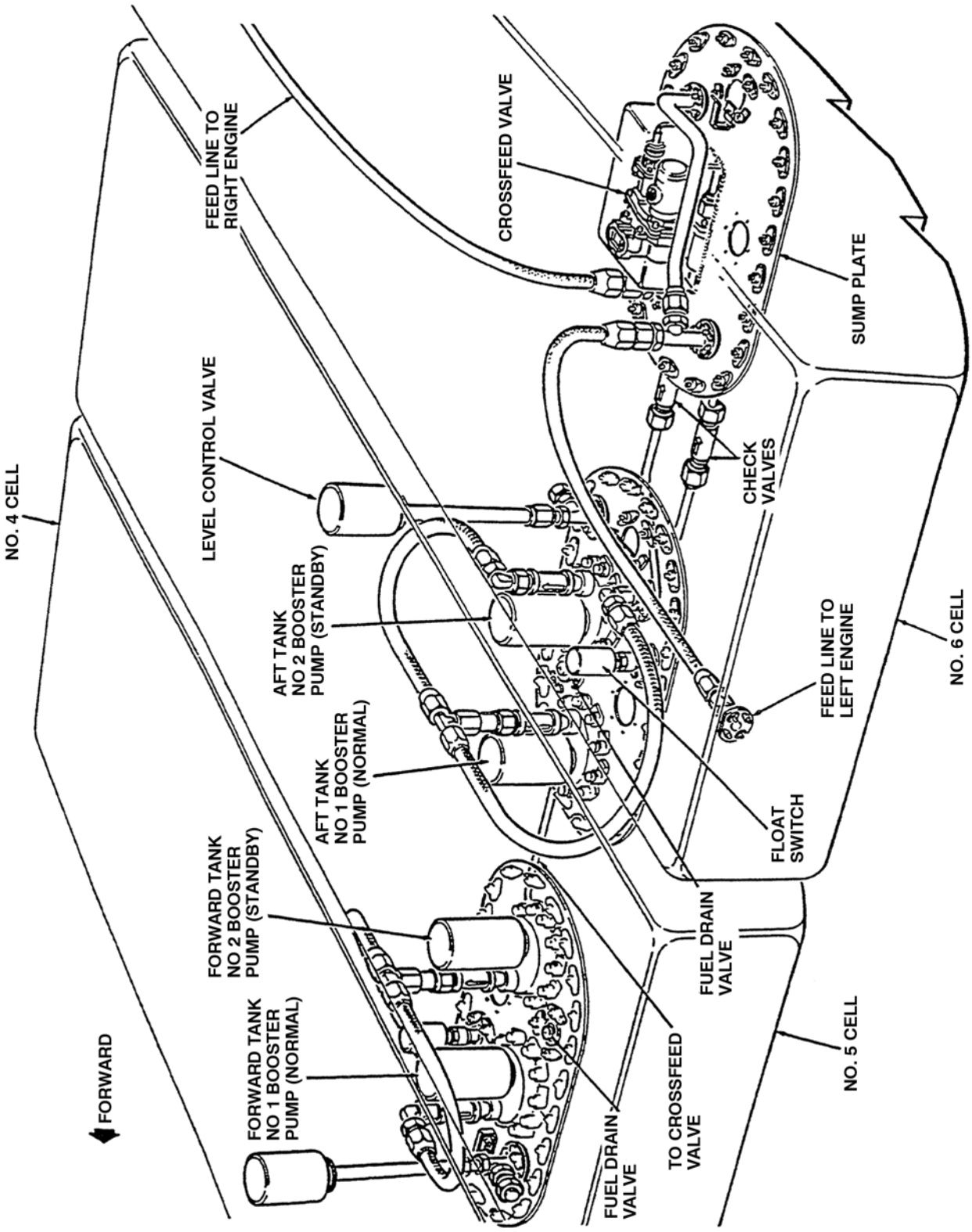
A. Fuel Booster Pumps

Four fuel booster pumps are installed, two in No. 4 cell of the forward tank and two in No. 5 cell of the aft tank. On aircraft with Mod 6/1795 installed, there are two submerged constant speed, centrifugal, electric fuel boost pumps and are located in each fuselage tank collector cell. The pumps deliver low-pressure fuel. They are powered from the left and right DC buses and are protected by circuit breakers labelled BOOST PUMP AFT 1, BOOST PUMP AFT 2, BOOST PUMP FWD 1 and BOOST PUMP FWD 2 on the main circuit breaker panel. Pre Mod 6/2267 pumps contain Captor Corp., part number A–3210 EMI filters installed in the circuit. The filters are mounted to the pump base. Post Mod 6/2267 pumps incorporate Lone Star, part number 122253–10A EMI filters installed in the circuit. The filters are mounted on a sheet metal plate installed on the RH Cabin sidewall at approximately station 135, below the window. On aircraft with Mod 6/2299 installed, improved fuel pumps have a new EMI/RFI electrical noise filter and ground circuit to improve corrosion tolerance of the pump and nearby components. Mod 6/2299 fuel pumps have two electrical fly leads that are connected to the aircraft wiring.

The fuel tank selector is located on the center instrument panel and marked FUEL SELECTOR. It is a rotary knob with three marked positions – BOTH ON FWD, NORMAL and BOTH ON AFT. In the NORMAL position the forward and aft tanks supply the right and left engines, respectively, the crossfeed valve is closed, and the boost pump(s) in each tank is energized. In BOTH ON FWD or AFT position, the crossfeed valve is open, the boost pumps of the non selected tank are automatically deenergized, and both boost pumps in the selected tank are energized (the boost pump switch selection is overridden). Boost pump switches located on the fuel panel are labelled BOOST PUMP 1 with ON, OFF and TEST positions marked. With power ON, circuit breakers set, and the FUEL SELECTOR positioned to NORMAL, the No. 1 boost pump in each collector tank is energized when ON position is selected; the No. 2 boost pumps remain inoperative. Control of the No. 2 boost pump in each tank is by means of an automatic electrical changeover sequence, which is initiated by the failure of the No. 1 boost pump. Refer to Figure 1 and Table 1 — Fuel Booster Pumps – Modification State.

Table 1 Fuel Booster Pumps – Modification State

MODIFICATION NO.	BOOSTER PUMP POSITION		TYPE OF PUMP	EMI FILTER
6/1795	No. 1	No. 2	Airborne (1D2–17)	A–3210 Captor Corp
6/2267	No. 1	No. 2	Airborne (1D2–20)	122253–10A Lone Star
6/2299	No. 1	No. 2	Airborne (1D2–21)	Embedded Filter (Captor Corp)



Booster Pumps and Crossfeed Valve Installation
Figure 1

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B. Crossfeed Valve

The crossfeed valve is an electric motor-operated gate valve which is controlled by the FUEL SELECTOR switch on the fuel control panel in the flight compartment. The valve is mounted in the fuel line on the floor of No. 6 fuel cell. To replace a crossfeed valve it will be necessary to first drain all front and rear fuel cells. Crossfeed indication is controlled by a switch located in the crossfeed valve assembly.

C. Fuel Emergency Shut-off Valves

There are two fuel emergency shut-off valves provided to shut-off the fuel supply in case of engine malfunction. Each valve is an electric motor-operated gate valve mounted in the nacelle behind the firewall. A switch is provided on the emergency panel for each valve and is marked FUEL EMERGENCY OFF and NORMAL. To replace an emergency shut-off valve, drain fuel from filter (refer to 28-20-31, Fuel Filter (Mod 6/2006) – Maintenance Practices) then remove valve.

D. Fuel Filter

A fuel filter is provided for each engine and is mounted behind the engine firewall to the rear of the fuel emergency shut-off valve. The fuel filter is an assembly consisting of a housing, filter element and bypass valve. Should the filter element become clogged, the fuel forces the bypass valve to open and the unfiltered fuel passes directly through the center of the filter. A drain valve is installed in the aft nacelle and connected to the filter by a hose.

E. Standby Booster Pump Emergency Switches

Two lever-lock standby booster pump emergency switches, marked BOOST PUMP 2 are located on the fuel control panel. The emergency switches override the automatic change-over system when selected to EMERGENCY to operate the corresponding No. 2 booster pump in the aft and/or forward fuselage tank. The switches are locked in the off position and must be pulled before a selection can be made. Refer to Figure 2.

F. Fuel Ejectors

The fuel ejectors are designed on the venturi principle so that no moving parts are required. Fuel under pressure from the main feed system passing through the ejector draws additional fuel from the tank and moves the total fuel through the transfer line into the main cell. There are two ejectors, one is mounted on the sump plate of No. 1 cell and the other on the sump plate of No. 8 cell.

G. Drain Valves

There are six water drain valves incorporated in the fuel system to provide a check for water accumulation. One is installed on each fuel filter and two in each front and rear fuel manifold. Those on the fuel filters are located in the engine nacelles, the manifold drains are located in the trough on the underside of the fuselage.

H. Fuel Level Control Valves

Fuel level control valves are installed on the sump plates of cells No. 4 (forward tank) and No. 5 (aft tank). The valves control the level of fuel in cells 4 and 5 by opening or closing the fuel transfer system line, and in cells 1 and 7 control the level of fuel in the forward and aft tanks by opening or closing the pressure refuel line.

I. Motive Line Check Valves

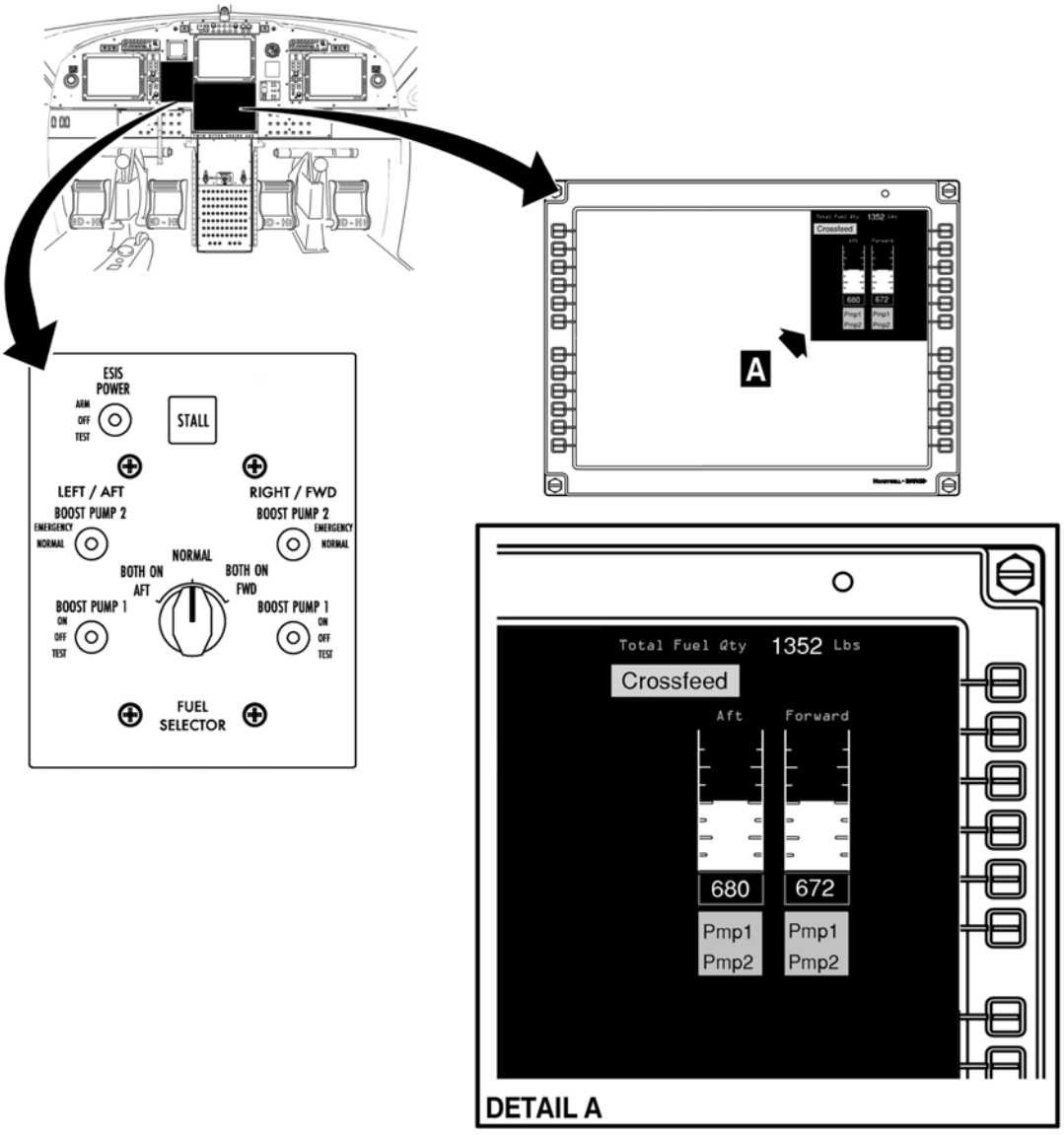
There are two motive line check valves fitted in the pressure lines to the ejectors, one beneath No. 1 cell and one beneath No. 7 cell. Each valve housing contains a fuel filter which can be removed for clearing when valve is disassembled.

J. Thermal Relief Check Valve

Two thermal relief check valves, one for each tank pressure line, are located below the forward edge of No. 6 fuel cell sump plate. The valve consists of a flap type check valve with an internal bleed provided to prevent thermal pressure build-up in the engine fuel supply lines.

K. Fuel Controls

Fuel controls consist of two main booster pump switches (BOOST PUMP 1), two emergency switches (BOOST PUMP 2), and a rotary FUEL SELECTOR switch. Refer to Figure 2.



Fuel Controls
Figure 2